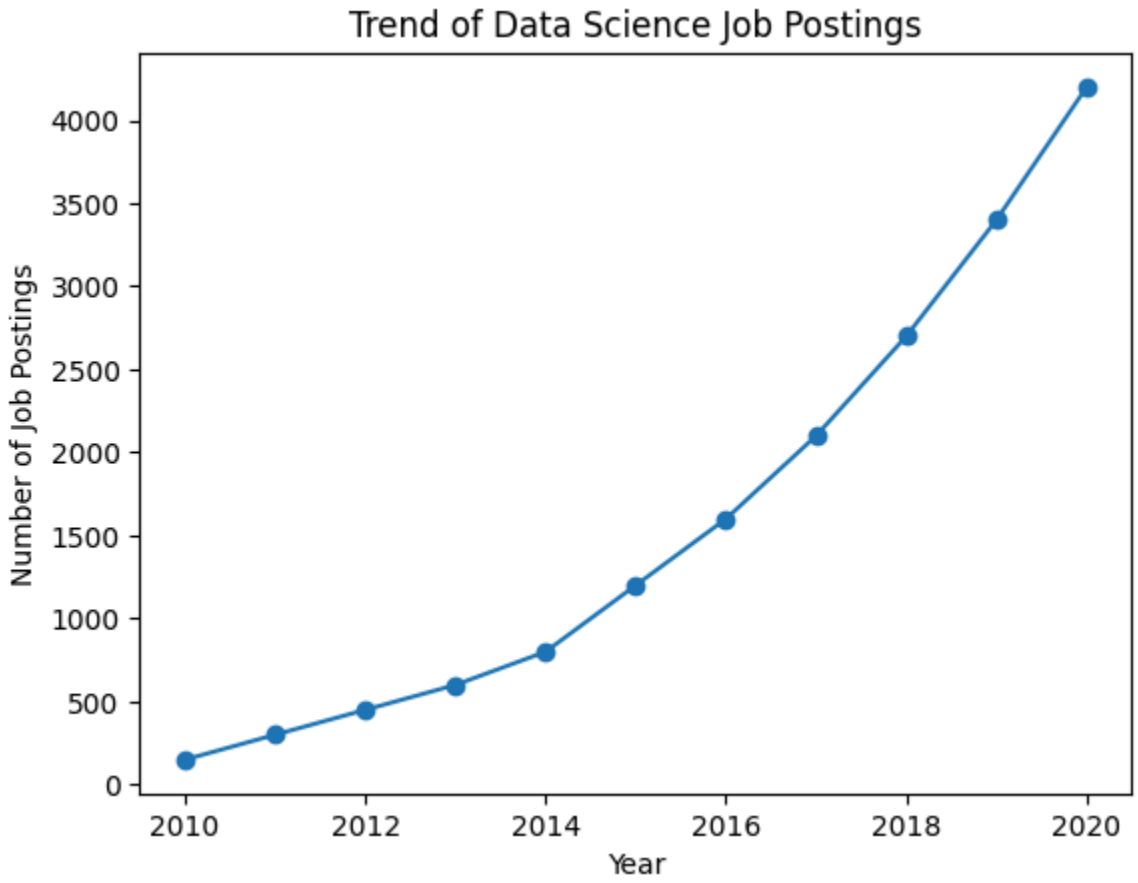
EXPERIMENT NO: 1(a)

Analyse the trend of data science job postings over the last decade

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
data={'Year': list(range(2010,2021)),
      'Job Postings': [150,300,450,600,800,1200,1600,2100,2700,3400,4200]}
df=pd.DataFrame(data)
plt.plot(df['Year'],df['Job Postings'],marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```

Output:



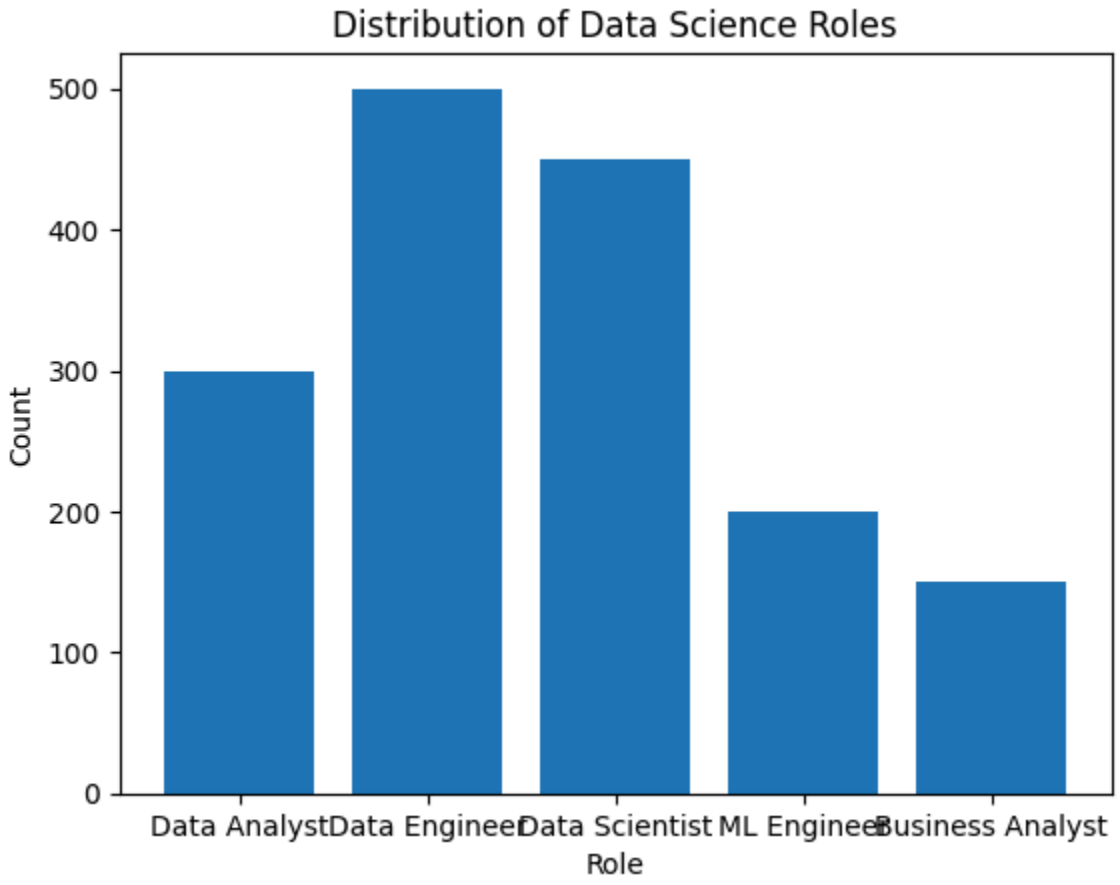
EXPERIMENT NO: 1(b)

Analyse and Visualize the Distribution of various Data Science Roles from a Dataset

Program:

```
import matplotlib.pyplot as plt
roles=['Data Analyst','Data Engineer','Data Scientist','ML Engineer',
'Business Analyst']
counts=[300,500,450,200,150]
plt.bar(roles,counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```

Output:



EXPERIMENT NO: 1(c)

Conduct an Experiment to differentiate datas based on data sets given

Program:

```
#Structured data
Structured_data=pd.DataFrame({
    'ID':[1,2,3],
    'Name':['Alice','Bob','Carlie'],
    'Age':[25,30,35]
})
print("Structured Data:\n",Structured_data)

#Unstructured data
unstructured_data="This is an example of Unstructured data. It can be a
piece of text, an image, or a video file."
print("\n Unstructured Data:\n",unstructured_data)

#Semi-structured data
semi_structured_data={'ID':1,'Name':'Alice','Attributes':{'Height':165,'We
ight':68}}
print("\nSemi-Structured Data:\n",semi_structured_data)
```

Output:

Structured Data:

	ID	Name	Age
0	1	Alice	25
1	2	Bob	30
2	3	Carlie	35

Unstructured Data:

This is an example of Unstructured data. It can be a piece of text, an image, or a video file.

Semi-Structured Data:

{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}

EXPERIMENT NO: 1(d)

Conduct an experiment to encrypt and decrypt given sensitive data

Program:

```
#Generate key and encrypt data
from cryptography.fernet import Fernet
key=Fernet.generate_key()
f=Fernet(key)
token=f.encrypt(b"Rajalakshmi Engineering College")
token
b'....'
f.decrypt(token)
b'Rajalakshmi Engineering College'
key=Fernet.generate_key()
cipher_suite=Fernet(key)
plain_text=b'Rajalakshmi Engineering College.'
cipher_text=cipher_suite.encrypt(plain_text)

#Decrpyt Data
decrypted_text=cipher_suite.decrypt(cipher_text)
print("Original Data: ",plain_text)
print("Encrpyted Data: ",cipher_text)
print("Decrpyted Data: ",decrypted_text)
```

Output:

Original Data: b'Rajalakshmi Engineering College.'

Encrpyted Data:

b'gAAAAABqoVa79_0dbI4tCrFzYT9-xcnJXcWvARfZYAHgo6lJmDPMYVvLf4dNCocedgX1zeUXov6H1EpfOsnQpP91NZwwbjORnRbXHlWP9CWeyiMBS8DFL0h-QG-X0vk3JHCsJ09btGUA'

Decrpyted Data: b'Rajalakshmi Engineering College.'